$$\frac{dy}{dx} = f(x,y), \quad y(x_0) = y_0.$$

$$f(x,y) = \frac{2(y-1)}{x}, \quad x_0 = 0, \; y(x_0) = 0.$$

discontinuous and solution does not exist.

$$\frac{dy}{dx} = f(x,y), \quad y(0) = 1.$$

discontinuous.

$y(x) = 1 + cx^2$, $c \to$ arbitrary constant.

infinitely many solution.

Lecture-5. Date 18.03.26.

Second order linear ODEs.

An ODE that can be written as.

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = r(x). \quad \text{——— (1)}$$

where, $p(x)$, $q(x)$, and $r(x)$ are functions defined on $I \subseteq \mathbb{R}$.

interval.

This is Called the Standard form of 2nd order linear ODE.

$$L \equiv \frac{d^2}{dx^2} + p(x)\frac{d}{dx} + q(x)\frac{d^0}{dx^0} \longrightarrow \text{Linear}.$$

$$: V \longrightarrow V.$$

collection of all functions whose all order derivatives exist.

(1) is called <u>homogeneous</u> if $r(x)=0$, otherwise (1) is called <u>non-homogeneous</u>.

<u>Example</u>: $\frac{d^2y}{dx^2}+25y=0$ 2nd order + linear + homogeneous.

$\frac{d^2y}{dx^2}+25y=e^{-x}\cos x$. 2nd order + linear + <u>non-homogeneous</u>.

<u>Example</u>. $\frac{d^2y}{dx^2}=y^2$, $\longrightarrow$ <u>non-linear</u>.

<u>Example</u>. $\frac{d^2y}{dx^2}+y=0$. —— (1).

$y_1=\sin x$, $y_2=\cos x$ are solutions of (1)

$y_3=\underbrace{A\sin x+B\cos x}$. is also a solution of (1).

This is called linear combination of $y_1=\sin x$ and $y_2=\cos x$.

<u>Theorem</u> (<u>Superposition/linearity Principle</u>)

For a homogeneous linear 2nd order ODE

$$\frac{d^2y}{dx^2}+p(x)\frac{dy}{dx}+q(x)\,y=0 \text{ —— } (*)$$

Any linear combination of solutions is

again a solution.

H.W. y_1, y_2 solution of (*).

$$y_3 = Ay_1 + By_2.$$

Remark: Superposition / linearity principle theorem holds only for homogeneous linear ODEs.

Example 1. (Non-homogeneous linear ODE)

$$y'' + y = 1. \quad \text{—— } (*)$$

$$y_1 = \sin x + 1 \qquad y_2 = \cos x + 1.$$

$y_3 = \sin x + \cos x + 2$ is not a solution of (*).

Example: (Non-linear ODE)

$$y\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0.$$

$y_1 = 1$, $y_2 = x^2$, $\rightarrow$ solutions:

but $y_3 = 1 + x^2$ is not a solution.

§ Initial Value Problem: (2nd order linear homogeneous ODE).

IVP:

$$\boxed{\begin{array}{l} y'' + p(x)y' + q(x)y = 0 \\ \qquad\qquad y(x_0) = k_0 \\ \qquad\qquad y'(x_0) = k_1 \end{array}} \quad \text{——(1)}$$

Existence and uniqueness Theorem.

If $p(x)$, $q(x)$ are continuous functions defined on some interval I containing x_0, then IVP (1) has a unique solution $y = y(x)$ defined on I.

Remark:- Every IVP has a Solution.

Definition:- Two solutions ϕ_1, ϕ_2 defined on I are called linearly dependent if $\exists\ (c_1, c_2) \neq (0,0)$ such that

$$c_1\phi_1 + c_2\phi_2 = 0 \quad \text{on } I.$$

i.e., $c_1\phi_1(x) + c_2\phi_2(x) = 0 \quad \forall x \in I.$

ϕ_1, ϕ_2 are called linearly independent if $c_1\phi_1 + c_2\phi_2 = 0$ on I, then $c_1 = 0 = c_2$.

(i.e., $c_1\phi_1(x) + c_2\phi_2(x) = 0 \quad \forall x \in I$).

Example: $\dfrac{d^2y}{dx^2} = y$., $y_1 = e^x$, $y_2 = e^{-x}$.

$$\left[\begin{array}{l} \text{So, } \dfrac{y_1}{y_2} = \dfrac{e^x}{e^{-x}} = e^{2x} \neq \text{constant.} \\ \text{So, } y_1, y_2 \text{ are linearly independent.} \end{array}\right]$$

$$c_1 y_1 + c_2 y_2 = 0.$$

$$\Rightarrow c_1 e^x + c_2 e^{-x} = 0.$$

$$c_1 e^x - c_2 e^{-x} = 0.$$

$$W(e^x, e^{-x}) = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2 \neq 0.$$

Wronskian.

§ W~~or~~nskian. of ϕ_1 and ϕ_2.

ϕ_1, ϕ_2 are defined on I, and ϕ_1', ϕ_2' exist on I.

$$W(\phi_1, \phi_2)(x) = \begin{vmatrix} \phi_1(x) & \phi_2(x) \\ \phi_1'(x) & \phi_2'(x) \end{vmatrix} \quad \forall x \in I.$$

Theorem:- Two solutions ϕ_1, ϕ_2 of IVP(1) defined on I are linearly independent if and only if $W(\phi_1, \phi_2)(x) \neq 0 \ \forall \ x \in I$.

<u>Proof</u>:- ($\Rightarrow$) Assume that ϕ_1, ϕ_2 are linearly independent solutions.

If possible, Suppose $W(\phi_1, \phi_2)(x_0) = 0$ for some $x_0 \in I$.

$$\begin{vmatrix} \phi_1(x_0) & \phi_2(x_0) \\ \phi_1'(x_0) & \phi_2'(x_0) \end{vmatrix} = 0.$$

Consider the equation

$$\phi_1(x_0) X_1 + \phi_2(x_0) X_2 = 0.$$

X_1, X_2 are variables.

$$\phi_1'(x_0) X_1 + \phi_2'(x_0) X_2 = 0.$$

There exists $(c_1, c_2) \neq (0,0)$ such that.

$$c_1 \phi_1(x_0) + c_2 \phi_2(x_0) = 0.$$
$$c_1 \phi_1'(x_0) + c_2 \phi_2'(x_0) = 0.$$

Set $\psi(x) = c_1 \phi_1(x) + c_2 \phi_2(x)$. $\forall x \in I$.

then. $\psi(x_0) = 0$

$\psi'(x_0) = 0$.

By uniqueness $\psi(x) = 0 \ \forall \ x \in I$.

i.e., $c_1\phi_1(x)+c_2\phi_2(x)=0 \quad \forall\ x\in I$.

$\Rightarrow$ ϕ_1, ϕ_2 are not linearly independant.

— a contradiction.

Hence, $W(\phi_1,\phi_2)(x)\neq 0 \quad \forall\ x\in I$.

($\Leftarrow$:)

$$c_1\phi_1+c_2\phi_2=0,$$

Since, $W(\phi_1,\phi_2)(x)\neq 0 \quad \forall\ x\in I$.

Fix $x_0\in I$,

$$\left.\begin{aligned} c_1\phi_1(x_0)+c_2\phi_2(x_0)&=0.\\ c_1\phi_1'(x_0)+c_2\phi_2'(x_0)&=0.\end{aligned}\right. \quad \left[\because W(\phi_1,\phi_2)(x_0)\neq 0\right]$$

$\Rightarrow$ $c_1=0=c_2$.

So, ϕ_1, ϕ_2 are linearly independent.

<u>Theorem</u>:- Let ϕ_1, ϕ_2 be two solutions of (1) defined on I. Let $x_0\in I$, then. ϕ_1 and ϕ_2 are linearly independent iff.

$$W(\phi_1,\phi_2)(x_0)\neq 0 \text{ for some } x_0\in I.$$

<u>Proof</u>:- H.W.

Lecture-6. Date 20.03.26.

Theorem Let ϕ_1, ϕ_2 be any two linearly independent solutions of

$$y'' + p(x)y' + q(x)y = 0. \quad \text{——(1)}$$

defined on I. Then every solution Φ of (1) is of the form

$$\Phi = c_1\phi_1 + c_2\phi_2, \text{ where } c_1, c_2 \text{ are constant.}$$

Proof:- Let $x_0 \in I$ be such that $W(\phi_1, \phi_2)(x_0) \neq 0$.

Let $\Phi(x_0) = \alpha$, $\Phi'(x_0) = \beta$.

Consider the system of equations

$$x_1\phi_1(x_0) + x_2\phi_2(x_0) = \alpha.$$
$$x_1\phi_1'(x_0) + x_2\phi_2'(x_0) = \beta.$$

Since $W(\phi_1, \phi_2)(x_0) \neq 0$, this system has unique solution c_1, c_2 such that

$$c_1\phi_1(x_0) + c_2\phi_2(x_0) = \alpha.$$
$$c_1\phi_1'(x_0) + c_2\phi_2'(x_0) = \beta.$$

Let $\psi(x) = c_1\phi_1(x) + c_2\phi_2(x) \quad \forall x \in I.$

Then $\psi(x_0) = \alpha$
$\psi'(x_0) = \beta$

By uniqueness of Solution for IVP (1).

$$\psi = \Phi$$

i.e., $\psi(x) = \Phi(x) \quad \forall \, x \in I$.

$\Rightarrow \quad \Phi(x) = c_1 \phi_1(x) + c_2 \phi_2(x) \quad \forall \, x \in I$.

$\square$.

§. <u>Homogeneous linear ODEs with constant coefficients:</u>

$$y'' + ay' + by = 0, \quad a, b \text{ constants.} \quad \text{—(1)}.$$

Let $y = e^{\lambda x}$.
$y' = \lambda e^{\lambda x}$.
$y'' = \lambda^2 e^{\lambda x}$.

$(\lambda^2 + a\lambda + b) e^{\lambda x} = 0$.

$\Rightarrow \underbrace{\lambda^2 + a\lambda + b = 0}_{\text{Characteristic equation of (1).}}$

$$\lambda = \frac{-a \pm \sqrt{a^2 - 4b}}{2}. \qquad M = \sqrt{a^2 - 4b}$$

① real & distinct. (if $a^2 > 4b$). $\lambda_1 = \frac{-a + M}{2}$, $\lambda_2 = \frac{-a - M}{2}$.

② real & equal (if $a^2 = 4b$), $\lambda_1 = -\frac{a}{2} = \lambda_2$.

③ complex, $a^2-4b<0$.

denote $R := \sqrt{4b-a^2}$.

$$\lambda_1 = \frac{-a+Ri}{2}$$

$$\lambda_2 = \frac{-a-Ri}{2}.$$

<u>Case-I</u>: $\lambda_1 = \frac{-a+M}{2}$, $\lambda_2 = \frac{-a-M}{2}$.

$y_1 = e^{\lambda_1 x}$, $y_2 = e^{\lambda_2 x}$.

$y_1/y_2 = \underbrace{e^{(\lambda_1-\lambda_2)x}}_{\text{not constant}}$. $[\because \lambda_1 \neq \lambda_2$.

So, y_1, y_2 are linearly independent.

<u>General solution</u>:

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}.$$

<u>Case-II</u> when $a^2=4b$, $\lambda_1 = -\frac{a}{2} = \lambda_2$.

Then $y_1 = e^{\lambda_1 x}$, $\frac{y_2}{y_1} = u$

$$y_2 = u y_1.$$

$$y_2' = u' y_1 + u y_1'$$

$$y_2'' = u'' y_1 + 2u' y_1' + u y_1''$$

$$y_1 u'' + (2y_1' + ay_1)u' + (y_1'' + ay_1' + by_1)u = 0$$

$$y_1 = e^{\lambda_1 x} \qquad (y_1'' + ay_1' + by_1) = 0. \quad [\because y_1 \text{ is a solution of (1)}].$$

$$2y_1' + ay_1 = \underline{(2\lambda_1 + a)} e^{\lambda_1 x}.$$
$$= 0.$$

$$\Rightarrow \quad y_1 u'' = 0 \Rightarrow u'' = 0 \Rightarrow u = c_1 x + c_2.$$

Take, $u = x$.

$$y_2 = x e^{\lambda_1 x}.$$

Then y_1, y_2 are linearly independent.

So, the general solution of (1). is

$$c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}; \quad c_1, c_2 \text{ are constants}.$$

<u>Case III</u>: $a^2 - 4b < 0$.

$$R = \sqrt{4b - a^2} > 0.$$

$$\lambda_1 = \frac{1}{2}(-a + iR).$$

$$\lambda_2 = \frac{1}{2}(-a - iR).$$

$$e^{\lambda_1 x} = e^{-\frac{a}{2}x}\left[\cos\left(\frac{R}{2}x\right) + i\sin\left(\frac{R}{2}x\right)\right].$$

$$\left.\begin{aligned} y_1 &= e^{-\frac{a}{2}x}\cos\left(\frac{R}{2}x\right) \\ y_2 &= e^{-a/2\,x}\sin\left(\frac{R}{2}x\right). \end{aligned}\right\}$$

H.W.

$W(y_1, y_2)(x) = R\,\frac{e^{-ax}}{2} \neq 0.$

So, y_1, y_2 are linearly independent.

Hence general solution of (1) is.

$$y = e^{-a/2\,x}\left[c_1 \cos\frac{R}{2}x + c_2 \sin\frac{R}{2}x\right]$$

Euler-Cauchy equation :

$$x^2y'' + axy' + by = 0, \quad \text{---(1)}$$

where a, b are constants.

take $\quad y = \phi(t). \qquad t = \ln x.$

$$\frac{dy}{dx} = \frac{d\phi}{dt}\cdot\frac{dt}{dx} = \frac{1}{x}\phi' \Rightarrow xy' = \phi'.$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\phi' + \frac{1}{x^2}\phi'' \Rightarrow x^2y'' = \phi'' - \phi'.$$

$$\phi'' - \phi' + a\phi' + b\phi = 0.$$

$$\Rightarrow \quad \phi'' + (a-1)\phi' + b\phi = 0.$$

$$\phi = e^{\lambda t} = e^{\lambda \ln x} = x^\lambda.$$

Let. $y = x^{\lambda}$ be a trial Solution of (1).

$y' = \lambda x^{\lambda-1}, \quad y'' = \lambda(\lambda-1)x^{\lambda-2}.$

$\lambda(\lambda-1)x^{\lambda} + a\lambda x^{\lambda} + b x^{\lambda} = 0.$

$\Rightarrow \left(\lambda(\lambda-1) + a\lambda + b\right)x^{\lambda} = 0.$

$\Rightarrow \lambda(\lambda-1) + a\lambda + b = 0.$

$\Rightarrow \lambda^2 + (a-1)\lambda + b = 0$

$$\lambda_1 = \frac{-(a-1) + \sqrt{(a-1)^2 - 4b}}{2}.$$

$$\lambda_2 = \frac{-(a-1) - \sqrt{(a-1)^2 - 4b}}{2}$$

Case:- I. $\lambda_1 \neq \lambda_2$.

$y_1 = x^{\lambda_1}$ and $y_2 = x^{\lambda_2}$.

Since. $\lambda_1 \neq \lambda_2$, y_1, y_2 are linearly independent.

So, general. Solution is.

$$y = c_1 x^{\lambda_1} + c_2 x^{\lambda_2}.$$

Case-II $(a-1)^2 - 4b = 0$. $\lambda_1 = -\frac{(a-1)}{2} = \lambda_2$.

check! $y_1 = x^{\lambda_1}$.

$y_2 = \ln x \, x^{\lambda_1}$

$\therefore$ general solution $y = (c_1 + c_2 \ln x)x^{\lambda_1}$.

Case-III :- H.W.